

SL No.X/Sem4 /I

Model Question Paper I

RENEWABLE ENERGY ENGINEERING-11

Time: 3 Hours

Max.Marks:75

PART A

Answer **all** questions in one word or one sentence. Each question carries **1 mark**.

1	What are the two Factors affecting distribution of wind energy on the surface of earth?	M1.01	R
2	Define Capacity factor of WECS.	M1.02	R
3	List two usable forms of Biomass.	M2.01	R
4	Define Flood Tide and Ebb Tide.	M3.02	R
5	Name any two wave energy conversion devices.	M3.03	R
6	List any two Tidal Technologies.	M3.02	R
7	Name any two types of OTEC power plants.	M3.04	R
8	Which are the two types of Liquid dominated Geothermal power plant?	M4.04	R
9	List any two applications of geothermal energy.	M4.03	R

PART B

Answer any **eight** questions from the following, each question carries **3 marks**.

(8 * 3=24 marks)

1	Illustrate the power versus wind speed characteristics .	M1.02	U
2	List any three advantages and limitations of HAWT.	M1.03	R
3	List any six advantages of Anaerobic digestion.	M2.03	R
4	Name three components of the Tidal power plant.	M3.02	R

5	List any six Limitations of Tidal power plants.	M3.02	R
6	Define Crest and Trough.	M3.03	R
7	Draw a Dolphin Type wave power machine and label the parts.	M3.03	R
8	Classify SHP depending on Capacity and Head.	M4.01	U
9	Show the block diagram of a Vapor dominated geothermal power plant.	M4.04	R
10	Illustrate Francis turbine with a neat sketch.	M4.02	U

PART C

Answer **ALL** questions. Each question carries **7 marks**.

(6*7=42 marks)

III	Draw and Explain the block diagram of wind energy conversion system	M1.01	U
	OR		
IV	Explain Constant speed Constant Frequency Scheme with block diagram	M1.04	U
V	A wind turbine travels at the speed of 10 m/s and has a blade length of 20 m, Air Density = 1.23 kg/m^3 . Determine wind power.	M1.02	A
	OR		
VI	Calculate the power converted from the wind into rotational energy in the turbine using the following data: Blade length=52m, Air Density = 1.23 kg/m^3 , power coefficient = 0.4, wind speed = 12m/sec.	M1.02	A
VII	Illustrate Hydrogen and Oxygen fuel cells and explain its working.	M2.04	U
	OR		
VIII	Summarize KVIC model of Biogas plant.	M2.03	U
IX	Explain the Biochemical method of Biomass conversion?	M2.02	U

	OR		
X	Draw and Explain Crossdraft Type biomass gasifiers .	M2.02	U
XI	Illustrate a closed cycle OTEC system with block diagram	M3.04	U
	OR		
XII	Explain the Double Basin arrangement of tidal power production	M3.02	U
XIII	Illustrate a liquid dominated flash steam Geothermal power plant	M4.04	U
	OR		
XIV	Explain the working of a bulb turbine with a neat sketch.	M4.02	U

Module Wise Question Analysis

Module	Hr/ module	(hi / ΣHi) * 123	Type of questions							
			Part A (1 mark)		Part B (3mark)		Part C (7mark)		Total	
			No	Mark s	No	Mark s	No	Mark s	No	Mark s
I	17	36.05	2		2		4		8	
			-----	2	-----	6	-----	28	-----	36
II	14	29.68	1		1		4		6	
			-----	1	-----	3	-----	28	-----	32
III	14	29.68	4		4		2		10	
			-----	4	-----	12	-----	14	-----	30
IV	13	27.56	2		3		2		7	
			-----	2	-----	9	-----	14	-----	25
Total			9		10		12		31	
			-----	9	-----	30	-----	84	-----	123

Cognitive Level Wise Question Analysis

level	% mark distribu tion	mark s/leve l	Type of questions							
			Part A (1 mark)		Part B (3mark)		Part C (7mark)		Total	
			No	Mark s	No	Mark s	No	Mark s	No	Mark s
R	30	36.9	9		7		0		16	
				9		21		0		30
U	50	61.5	0		3		10		13	
				0		9		70		79
A	20	24.6	0		0		2		2	
				0		0		14		14
Total			9		10		12		31	
				9		30		84		123

Question Wise Analysis

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.01	R	1	2
I.2	M1.02	R	1	2
I.3	M2.01	R	1	2
I.4	M3.02	R	1	2
I.5	M3.03	R	1	2
I.6	M3.02	R	1	2
I.7	M3.04	R	1	2
I.8	M4.04	R	1	2
I.9	M4.03	R	1	2
II.1	M1.02	U	3	8
II.2	M1.03	R	3	7
II.3	M2.03	R	3	7
II.4	M3.02	R	3	7
II.5	M3.02	R	3	7
II.6	M3.03	R	3	7
II.7	M3.03	R	3	7
II.8	M4.01	U	3	8
II.9	M4.04	R	3	7
II.10	M4.02	U	3	8
III.	M1.01	U	7	17
IV.	M1.04	U	7	17
V	M1.02	A	7	17

VI	M1.02	A	7	17
VII	M2.04	U	7	17
VIII	M2.03	U	7	17
IX	M2.02	U	7	17
X	M2.02	U	7	17
XI	M3.04	U	7	17
XII	M3.02	U	7	17
XIII	M4.04	U	7	17
XIV	M4.02	U	7	17
Total			123	295

Prepared By :



Mrs. Fathima Salim
Assistant Professor
EEE Dept.
College of Engineering Pathanapuram

Scrutinized By :

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Aravind M S
Lecturer in Electrical Engineering
Central Polytechnic College Vattiyoorkavu, Thiruvananthapuram

SL No.X/Sem4/II

Model Question Paper II

RENEWABLE ENERGY ENGINEERING-11

Time: 3 Hours

Max.Marks: 75

PART A

Answer **all** questions in one word or one sentence. Each question carries **1 mark**.

1	List any two schemes for wind power generation.	M1.04	R
2	Name any two applications of wind power.	M1.02	R
3	List any two applications of Fuel cell.	M2.04	R
4	What are the two Ocean Energy Resources?	M3.01	R
5	Define Tide.	M3.02	R
6	Define Tidal Range.	M3.02	R
7	List two limitations of wave energy.	M3.03	R
8	What are Magma Resources?	M4.03	R
9	Name any two water turbines used in SHP.	M4.02	R

PART B

Answer any **eight** questions from the following, each question carries **3 marks**.

(8 * 3=24 marks)

1	List any three Limitations of WECS	M1.01	R
2	List the classification of WECS	M1.01	R
3	What is the difference between Battery and Fuel cell	M2.04	R
4	Explain the basic principle of OTEC	M3.04	U
5	Show the block Diagram of a Hybrid cycle OTEC system	M3.04	R
6	Explain High level Reservoir Wave Machine	M3.03	U

7	Show the schematic layout of Oscillating Water Column	M3.03	R
8	List the nature of geothermal fields	M4.03	R
9	Explain Kaplan Turbine	M4.02	U
10	What is Hot Dry Rock Resources	M4.03	R

PART C

Answer **ALL** questions. Each question carries **7 marks**.

(6*7=42 marks)

III	A wind turbine travels at the speed of 20 m/s and has a blade length of 50 m, Air Density = 1.23 kg/m^3 . Determine wind power	M1.02	A
	OR		
IV	Derive the expression for power available from the wind	M1.02	A
V	Compare Horizontal Axis and Vertical Axis Wind Turbines	M1.03	U
	OR		
VI	Explain the characteristics and advantages of Darrieus Rotor	M1.03	U
VII	Draw and Explain Updraft Type biomass gasifiers .	M2.02	U
	OR		
VIII	Illustrate JANATA model of Biogas plant	M2.03	U
IX	Explain the Pyrolysis and Liquefaction process	M2.02	U
	OR		
X	Summarize the continuous and Batch type Biogas Plant	M2.03	U
XI	Illustrate Open cycle OTEC system with block diagram	M3.03	U
	OR		
XII	Summarize the Single Basin arrangement of tidal power production	M3.04	U

XIII	Explain a liquid dominated Binary cycle Geothermal power plant	M4.04	U
	OR		
XIV	Draw and explain the layout of Small Hydro Scheme	M4.01	U

Module Wise Question Analysis

Module	Hr/ modul e	(hi / ΣHi) * 123	Type of questions							
			Part A (1 mark)		Part B (3mark)		Part C (7mark)		Total	
			No	Mar ks	No	Mar ks	No	Mar ks	No	Mar ks
I	17	36.05	2		2		4		8	
			-----	2	-----	6	-----	28	-----	36
II	14	29.68	1		1		4		6	
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III	14	29.68	4		4		2		10	
			-----	4	-----	12	-----	14	-----	30
IV	13	27.56	2		3		2		7	
			-----	2	-----	9	-----	14	-----	25
Total			9		10		12		31	
			-----	9	-----	30	-----	84	-----	123

Cognitive Level Wise Question Analysis

level	% mark distribution	marks/level	Type of questions							
			Part A (1 mark)		Part B (3mark)		Part C (7mark)		Total	
			No	Marks	No	Marks	No	Marks	No	Marks
R	30	36.9	9		7		0		16	
				9		21		0		30
U	50	61.5	0		3		10		13	
				0		9		70		79
A	20	24.6	0		0		2		2	
				0		0		14		14
Total			9		10		12		31	
				9		30		84		123

Question Wise Analysis

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.04	R	1	2
I.2	M1.02	R	1	2
I.3	M2.04	R	1	2
I.4	M3.01	R	1	2
I.5	M3.02	R	1	2
I.6	M3.02	R	1	2
I.7	M3.03	R	1	2
I.8	M4.03	R	1	2
I.9	M4.02	R	1	2
II.1	M1.01	R	3	7
II.2	M1.01	R	3	7
II.3	M2.04	R	3	7
II.4	M3.04	U	3	8
II.5	M3.04	R	3	7
II.6	M3.03	U	3	8
II.7	M3.03	R	3	7
II.8	M4.03	R	3	7
II.9	M4.02	U	3	8
II.10	M4.03	R	3	7
III.	M1.02	A	7	17
IV.	M1.02	A	7	17

V	M1.03	U	7	17
VI	M1.03	U	7	17
VII	M2.02	U	7	17
VIII	M2.03	U	7	17
IX	M2.02	U	7	17
X	M2.03	U	7	17
XI	M3.03	U	7	17
XII	M3.04	U	7	17
XIII	M4.04	U	7	17
XIV	M4.01	U	7	17
Total			123	295

Prepared By :



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Assistant Professor
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SIGNATURE

A handwritten signature in blue ink, appearing to be 'Aravind M S', with a long horizontal stroke extending to the right.

Aravind M S

Lecturer in Electrical Engineering

Central Polytechnic College Vattiyoorkavu, Thiruvananthapuram

Scoring Indicators

Model Question Paper I

RENEWABLE ENERGY ENGINEERING-11

Q No	Scoring Indicators	Split score	Sub Total	Total Score
	PART A			9
I. 1	Factors affecting distribution of wind energy on the surface of earth <i>list any two</i> <i>2* 0.5 marks = 1 mark</i>	0.5+ 0.5	1	
	● Wind speed and direction ● Air temperature ● Air pressure			
I. 2	Capacity factor of WECS <i>1 mark</i>	1	1	
	Capacity factor = Annual Energy Output/(Rated power*Time in a year)			
I. 3	Two usable form of Biomass <i>list-any two</i> <i>2* 0.5 marks = 1 mark</i>	0.5+0.5	1	
	● Fuel wood ● Charcoal ● Bio-Diesel ● Bio-ethanol ● Biogas.			
I. 4	Flood Tide and Ebb Tide Definition <i>0.5mark each</i> <i>2* 0.5 marks = 1 mark</i>	0.5+ 0.5	1	
	Ebb is the tidal phase during which the tidal current is flowing seaward (ebb current) and flood is the tidal phase during which the tidal current is flowing inland (flood current);			

I. 5	Two wave energy conversion devices <i>0.5mark each</i> <i>2* 0.5 marks = 1 mark</i>	0.5+ 0.5	1	
	1) High level Reservoir Wave machine 2) Dolphin Type Wave power Machine			
I. 6	Two Tidal Technologies <i>0.5mark each</i> <i>2* 0.5 marks = 1 mark</i>	0.5+0.5	1	
	1)Tidal stream technologies 2)Tidal range technologies			
I. 7	Two types of OTEC power plants <i>0.5mark each</i> <i>2* 0.5 marks = 1 mark</i>	0.5+0 .5	1	
	1.Open Cycle OTEC 2.Closed Cycle OTEC			
I. 8	Two types of Liquid dominated Geothermal power plant <i>0.5mark each</i> <i>2* 0.5 marks = 1 mark</i>	0.5+0 .5	1	
	1.Flash steam system 2.Binary Cycle System			
I. 9	List two applications of Geothermal energy <i>Any two</i> <i>0.5*2 =1</i>	1	1	
	• Space heating and cooling. • Generation of electrical power. • Industrial process heat.			
	PART B			24
II. 1	Power versus wind speed characteristics <i>figure: 1 .5mark</i> <i>explanation-1.5 marks</i>	1.5+ 1.5	3	

	Wind power curve Wind power is exponentially proportional to wind speed. If wind speed doubles, power generation becomes eight times greater. Wind speed largely determines the amount of electricity generated by a turbine. Higher wind speeds generate more power because stronger winds allow the blades to rotate faster. Faster rotation translates to more mechanical power and more electrical power from the generator			
II. 2	List three advantages and limitations <i>Each point -0.5mark</i> <i>Total-3marks</i>	0.5*6	3	
	Advantages • It has a self-starting ability. It does not require any external power source to start. • Capable of working in high wind speed conditions. • In the case of slow wind conditions, its angle of attack can be varied to get maximum possible efficiency. • Since all blades of this turbine work simultaneously, it is capable of extracting maximum energy from the wind.			

	Disadvantages • Its initial installation cost is high. • It requires a large ground area for its installation. • Because of its giant size of blades and towers, it becomes difficult to transport it to the sites. • High maintenance cost. • Creates noise problems. • It cannot be installed near the human population. • It is not good for the bird's population. They are killed by its blades rotation.			
II. 3	List six advantages of Anaerobic digestion <i>Each point; 0.5 mark</i>	0.5*6	3	
	Advantages of anaerobic digestion. There are number of advantages of anaerobic digestion. 1. Calorific value of gas. One of the main benefits is the production of a bioproduct the biogas which has a calorific value and can therefore, be used as an energy source to produce steam or hot water. Because in dairy industries energy source is very important for dairy use, so there is no problem of gas storage or supply, but gas can be directly useful in heat energy. 2. New sludge production. The conversion of organic matter to methane and carbon dioxide results in a smaller quantity of excess sludge. 3. Stable sludge. In the case of municipal digestion the main reason for their installation was to produce a non-putrescable and inoffensive sludge and in many cases only a proportion of the gas produced was utilised. 4. Low running cost. There is no aeration in the anaerobic treatment naturally in this digestion, running costs are a quarter of the equivalent aerobic system. 5. Low odour. Since the system is enclosed the odours are contained. Compounds which are responsible for odour are broken down during digestion. The only slight odour of hydrogen sulphide normally presents in gas. However if the gas is burnt the problem will not arise. 6. Stability. A well adapted anaerobic sludge can be presented unfed for a considerable period of time without appreciable deterioration.			
II. 4	Any three components of Tidal power plant <i>Each component-1marks</i>	3	3	

	• powerhouse, • dam to form the basin, • Sluiceways from the basin to sea and vice versa. The function of the dam is to create an obstacle between the sea and the basin. The sluiceways are used to either fill the basin during high tide or empty the basin during the low tide.			
II. 5	Limitations of Tidal power plant <i>Any six limitations:0.5 marks for each limitations</i>	$0.5 \times 6 = 3$	3	
	• The capital cost of the plant is high and needs a long construction time. • The interest and depreciation component of the generation cost is very high. • The construction of tidal barrage calls for advanced technology due to very complex problems associated with it. The regions which are most suitable for tidal energy generation tend to be the ones which are difficult for barrage construction. So every location is not ideal for installing such a system. • Tidal power output is not constant throughout the day (unless pumping is used). The production is the form of pulses (2 pulses every day and each pulse lasting about 4 to 6 hours). The extent to which the power system can absorb the tidal pulses depends on the static and dynamic characteristics of the system. • Sedimentation of basins is a problem. • Seawater is very corrosive in nature. Marine life is affected			
II. 6	Crest and Trough Definition <i>Each definition:1.5 marks</i>	$1.5 + 1.5$	3	

	A crest point is a point on a wave with the maximum value of upward displacement within a cycle. A crest is a point on a surface wave where the displacement of the medium is at a maximum. A trough is the opposite of a crest, so the minimum or lowest point in a cycle.			
II. 7	Dolphin Type wave power machine <i>Figure with correct labeling-3 marks</i>	3	3	
II. 8	SHP classification depending on Capacity and Head <i>Based on capacity: 1.5 marks</i> <i>Based on Head:1.5 marks</i>	1.5+1.5	3	

1. Depending on Capacity

Size	unit size	Installation
Micro	upto 100 kW	100 kW
Mini	101 to 1000 kW	2,000 kW
Small	1001 to 6000 kW	15,000 kW

2. Depending on Head

Ultra low head	Below 3 metres
Low head	less than 30 metres
Medium head	Between 30 to 75 metres
High head	Above 75 metres.

II. 9

Block diagram of Vapor dominated geothermal power plant

Block diagrams: 3 marks

3

3

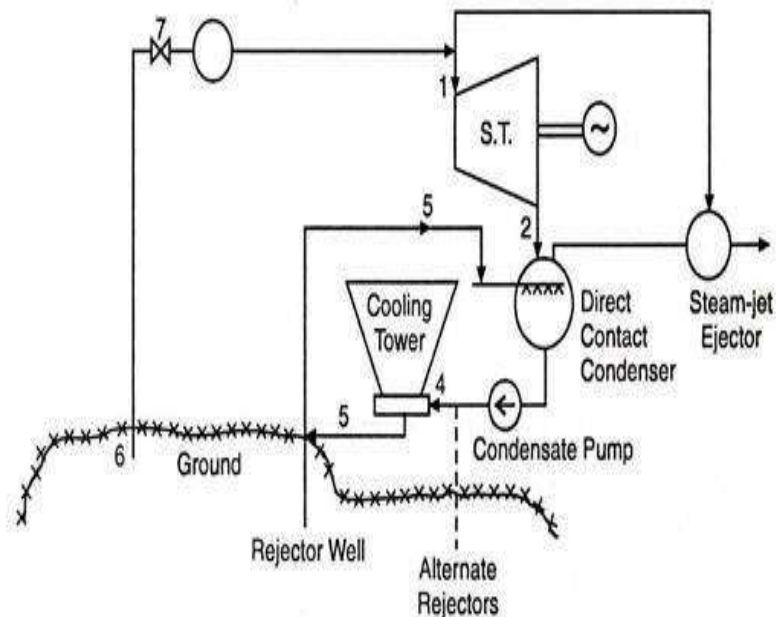
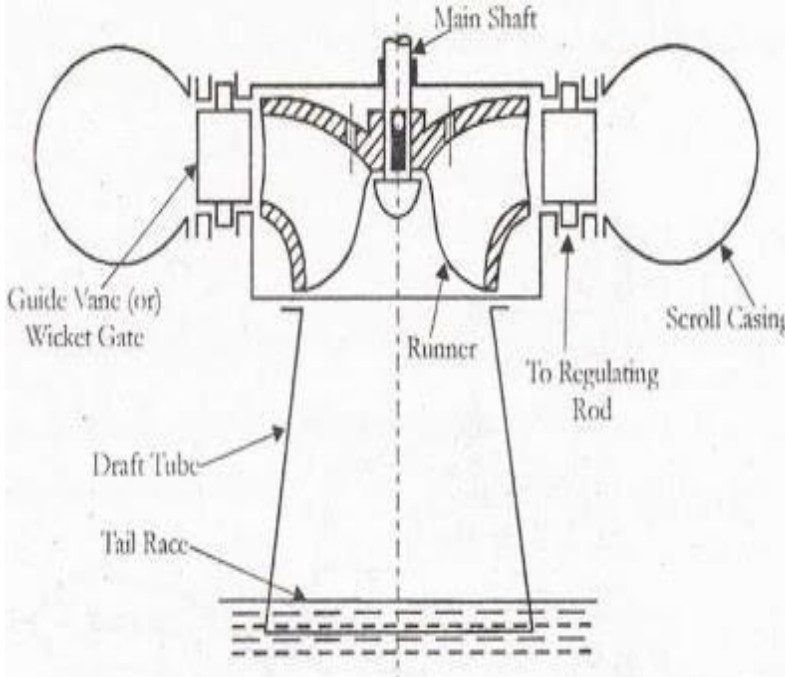
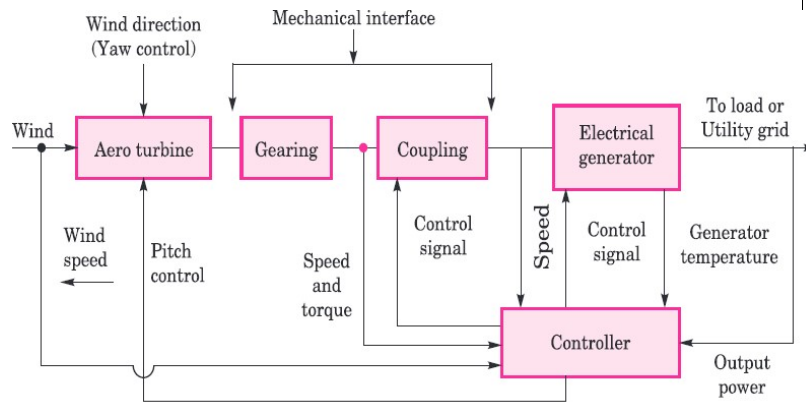


Fig. 7.1. Vapour-dominated Power Plant

II. 10	Francis turbine <i>Figure: 1.5 marks</i> <i>Explanation: 1.5 marks</i>	1.5+ 1.5	3	
	It is a reaction turbine. In reaction turbines the water entering the runner possesses pressure energy and this water in turn does work on the vanes by the principle of reaction. Fig shows a simple schematic representation of a Francis turbine, which is the mostly used reaction turbine. It consists of an inner rotating vanes forming the runner, surrounded by an outer ring of stationary guiding mechanism. Water from the penstock which is the pipe connecting the reservoir and the turbine flows into a scroll casing surrounding the turbine runner. From the scroll casing water flows through the guiding mechanism and enters the runner. After imparting energy to the runner, the water discharges through a draft tube. 			
	PART C			42
III	Block diagram of WECS <i>Block diagram: 3 marks</i> <i>Explanation: 4 marks.</i>	3+4	7	

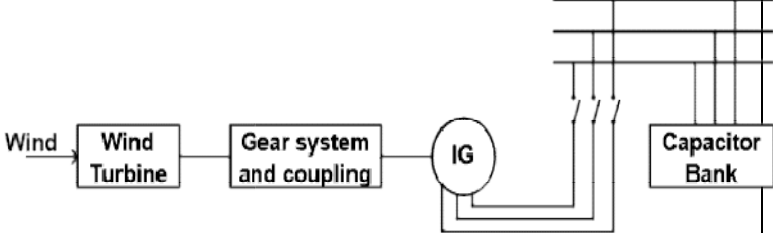


Aero turbines convert energy in wind to rotational mechanical energy. They require pitch control and yaw control (only in case of horizontal or wind axis machines) for proper operation. Mechanical Interface consisting of a step-up gear and a suitable coupling converts the rotational mechanical energy to an electrical energy, by use of an electrical generator. The output of this generator is connected to the load or power grid, as per the requirement

Yaw Control :

For localities, with the prevailing wind in one direction, the design of the turbine is simplified, where in the rotor is in fixed orientation with the swept area perpendicular to the predominant wind direction. Such a machine is said to be "Yaw fixed". Most wind turbines are yaw active, i.e. as the wind direction changes, a motor rotates the turbine slowly about the vertical axis (or yaw) so as to face the blades into the wind. The area of the wind stream swept by the wind rotor is then maximum.

In small turbines, yaw action is controlled by a tall vane. In larger machines a servo mechanism operated by a wind direction sensor controls the yaw motor that keeps the turbines properly oriented. The purpose of the controller is to sense wind speed, wind direction, shaft speed and torque at one or more points, power output and generator temperature as necessary and appropriate control signals for matching the electrical output to the wind energy input and protect the system from extreme conditions brought upon by strong winds, electrical faults etc.

IV	Constant speed Constant Frequency Scheme <i>Block diagram: 3 marks</i> <i>Explanation :4 marks</i>	4+3	7	
	 Constant speed drive is used for large generators that feed the generated power to the grid. Commonly Synchronous generators or Squirrel cage Induction generators are used for power generation (a) Synchronous Generator. For such machines the requirement of constant speed is very rigid and only minor fluctuations about 1% for short durations (fraction of a second) could be allowed. Synchronisation of wind driven generator with power grid also will pose problems with gusty winds. (b) Induction Generator. If the stator of an induction machine is connected to the power grid and if the rotor is driven above synchronous speed N_s ($N_s = 120 f/p$), the machine becomes a generator and delivers constant line frequency power to the grid, (f = line frequency and p = number of poles for which the stator winding is made). The per unit slip is 0 and 0.05. The output power of wind driven induction generator is uniquely determined by the operating speed. The pull out Torque should not be exceeded			
V	<u>Wind power calculation</u> <i>Area : 2marks</i> <i>Available power: 5marks</i>	5 + 2	7	

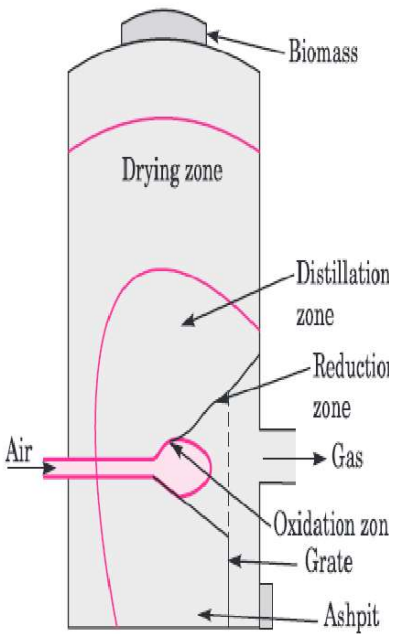
	- Given: - Wind speed $v = 10 \text{ m/s}$, - Blade length $l = 20 \text{ m}$, - air density $\rho = 1.23 \text{ kg/m}^3$, - area, $A = \pi r^2$ $= \pi \times 400$ $= 1256 \text{ m}^2$ - The wind power formula is given as, $P = \frac{1}{2} \rho A V^3$ $= 0.5 \times 1.23 \times 1256 \times 1000$ $P = 772440 \text{ W}$.			
VI	<u>Converted power from the wind</u> <i>Area : 2marks</i> <i>Available power: 5marks</i>	5+2	7	
	$l = r = 52 \text{ m}$ area, $A = \pi r^2$ $= \pi \times 52 \times 52 = 8495 \text{ m}^2$ $P = \frac{1}{2} \rho A V^3 C_p$ $= 0.5 \times 1.23 \times 8495 \times 0.4 \times 12^3$ $= 3.6 \text{ MW}$			
VII	Hydrogen Oxygen fuel cell <i>Figure-3 marks</i> <i>Explanation: 4 mark</i>	3+4	7	

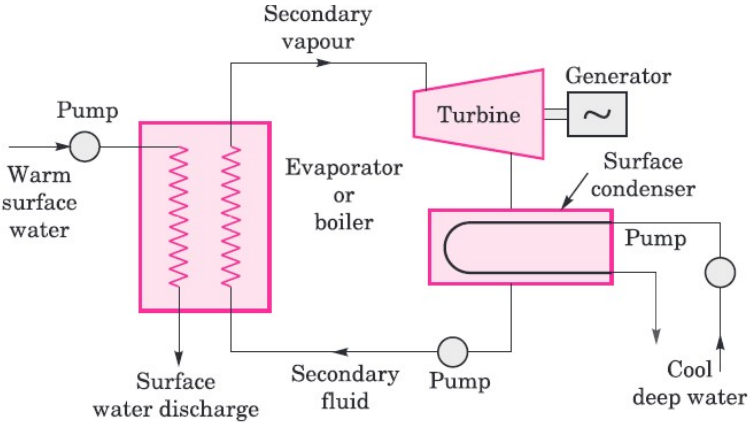
	A fuel cell is a lot like a battery. It has two electrodes where the reactions take place and an electrolyte which carries the charged particles from one electrode to the other. In order for a fuel cell to work, it needs hydrogen (H₂) and oxygen (O₂). The hydrogen enters the fuel cell at the anode. A chemical reaction strips the hydrogen molecules of their electrons and the atoms become ionized to form H⁺. The electrons travel through wires to provide a current to do work. The oxygen enters at the cathode, usually from the air. The oxygen picks up the electrons that have completed their circuit. The oxygen then combines with the ionized hydrogen atoms (H⁺), and water (H₂O) is formed as the waste product which exits the fuel cell. The electrolyte plays an essential role as well. It only allows the appropriate ions to pass between the anode and cathode. If other ions were allowed to flow between the anode and cathode, the chemical reactions within the cell would be disrupted			
	The reactions involved in a fuel cell are as follows: Anode side (an oxidation reaction): $2\text{H}_2 \Rightarrow 4\text{H}^+ + 4\text{e}^-$ Cathode side (a reduction reaction): $\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \Rightarrow 2\text{H}_2\text{O}$ Net reaction (the "redox" reaction): $2\text{H}_2 + \text{O}_2 \Rightarrow 2\text{H}_2\text{O}$			
VIII	KVIC model of Biogas plant <i>Diagram: 3.5 marks</i> <i>Description: 3.5 marks</i>	3.5+ 3.5	7	

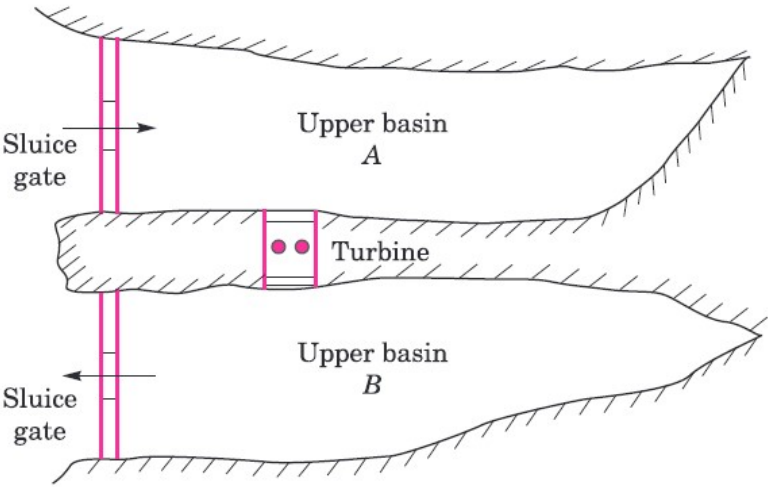
	This mainly consists of a digester or pit for fermentation and a floating drum for the collection of gas. Digester is 3.5-6.5 m in depth and 1.2 to 1.6 m in diameter. There is a partition wall in the center, which divides the digester vertically and submerges in the slurry when it is full. The digester is connected to the inlet and outlet by two pipes. Through the inlet, the dung is mixed with water (4:5) and loaded into the digester. The fermented material will flow out through the outlet pipe. The outlet is generally connected to a compost pit. The gas generation takes place slowly and in two stages. In the first stage, the complex, organic substances contained in the waste are acted upon by a certain kind of bacteria, called acid formers and broken up into small-chain simple acids. In the second stage, these acids are acted upon by another kind of bacteria, called methane formers and produce methane and carbon dioxide. The gas holder is a drum constructed of mild steel sheets. This is cylindrical in shape with concave. The top is supported radially with angular iron. The holder fits into the digester like a stopper. It sinks into the slurry due to its own weight and rests upon the ring constructed for this purpose. When gas is generated the holder rises and floats freely on the surface of slurry. A central guide pipe is provided to prevent the holder from tilting. The holder also acts as a seal for the gas. The gas pressure varies between 7 and 9 cm of water column. Under shallow water table conditions, the adopted diameter of digester is more and depth is reduced. The cost of the drum is about 40% of the total cost of the plant. It requires periodical maintenance.			
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IX	Biochemical method of Biomass conversion <i>2 methods: 3.5 marks each</i>	3.5+3 .5	7	

	<i>Biochemical conversion takes two forms. Anaerobic digestion and fermentation. Anaerobic digestion involves the microbial digestion of biomass. (An anaerobe is a micro-organism that can live and grow without air or oxygen, it gets its oxygen by the decomposition of matter containing it). It has already been used on animal manure but it also possible with other biomass. The process takes place at low temperature upto 65°C, and requires a moisture content of at least 80 per cent. It generates a gas consisting mostly of CO₂ and methane (CH₄) with minimum impurities such as hydrogen sulfide. The gas can be burned directly or upgraded to synthetic natural gas by removing the CO₂ and the impurities. The residue may consists of protein rich sludge that can be used as animal feed and liquid effluents that are biologically treated by standard techniques and returned to the soil.</i> Fermentation is the breakdown of complex molecules in organic compound under the influence of a ferment such as yeast, bacteria, enzymes, etc. Fermentation is a well-established and widely used technology for the conversion of grains and sugar crops into ethanol. About 500 million gal ethanol per year by 1985, were produced in the limited states by the use of surplus gain. It is intended for mixing with gasoline to produce <i>gasohol</i> (90% gasoline, 10% ethanol). This process requires high cost and high energy required. One scheme considered for reducing costs of ethanol production by fermentation is in finding less expensive grains or sugars and a process that requires less energy. Glucose produced by hydrolysis of an abundant carbohydrate polymer called <i>ligno cellulose</i> is being considered for the former.			
X	Cross draft Type biomass gasifiers <i>figure:3.5 marks</i> <i>explanation:3.5 marks</i>	3.5+3.5	7	

	(iii) Cross Draught Gasifiers. In this type of gasifiers, the gas produced passes upwards in the annular space around the gasifier that is filled with charcoal. The charcoal acts as an insulator and a dust filter. They are usually suitable for power generation upto 50 kW. Air enters the cross draft gasifier through a water cooled nozzle mounted on one side of the firebox. The gas is produced in the horizontal zone in front of the nozzle and passes through a vertical grate into the hot gas port on the opposite side. Because of the short path-length for the gasification reactions, this type of gas producer responds most rapidly for changes in gas production. However, the cross draft gasifier is not commonly used. Understanding of the various complex reactions occurring in a gasifier has so far been incomplete. However, the following reactions explain fairly well the process of gasification.	 Fig. 7.28. Schematic diagram of a crossdraft gasifier.		
XI	Closed cycle OTEC system. <i>Diagram: 3.5 marks</i> <i>Description: 3.5 marks</i>	3.5+ 3.5	7	
	In this system, working fluids having a low boiling point like ammonia, Freon-12, butane gas are used for heat engines because the working temperatures of sea water are small. The schematic diagram of a Closed OTEC power plant is shown in Figure.			

	The warm water from the surface of the sea is circulated through a heat exchanger with the help of a pump. In the heat exchanger, the heat of seawater is absorbed by Freon and Freon vapors are generated at high pressure. This vapor expands in the turbine to drive it and a coupled generator with it. The Freon vapor from the turbine condensed in a condenser with the help of cold water. The Freon condensate is pumped again into the heat exchanger and the complete cycle is repeated.			
				
XII	Double Basin arrangement <i>Diagram: 3.5 marks</i> <i>Description: 3.5 marks</i>	3.5+ 3.5	7	

	In a double basin tidal power plant, two basins at different levels are made, and a dam is provided in between them. Inlet and outlet sluice gates are made in the dam. The water level in the upper basin is maintained above the level of water in the lower basin. The upper basin is filled with water during high tide, and the lower basin is evacuated during the low tide. Therefore, a permanent head is produced between the upper and lower basins. When a sufficient head is created; the turbines of the power plant are started. The water flows from the upper basin to the turbine, which discharges into the lower basin, and electrical power is generated. When the water level in the upper basin is maximum during high tide, the inlet sluice is closed, and the level of the water in the lower basin keeps on rising due to the discharge of water by the turbine. When the level of water in the lower basin equals during the low tide, the outlet sluice is opened, and it is closed when the water level reaches up to its minimum level. In this system, the power can be produced continuously during the emptying and filling of basins.			
				
XIII	Liquid dominated flash steam Geothermal power plant <i>Diagram-- 3.5 marks</i> <i>Explanation- 3.5 marks</i>	3.5 + 3.5	7	

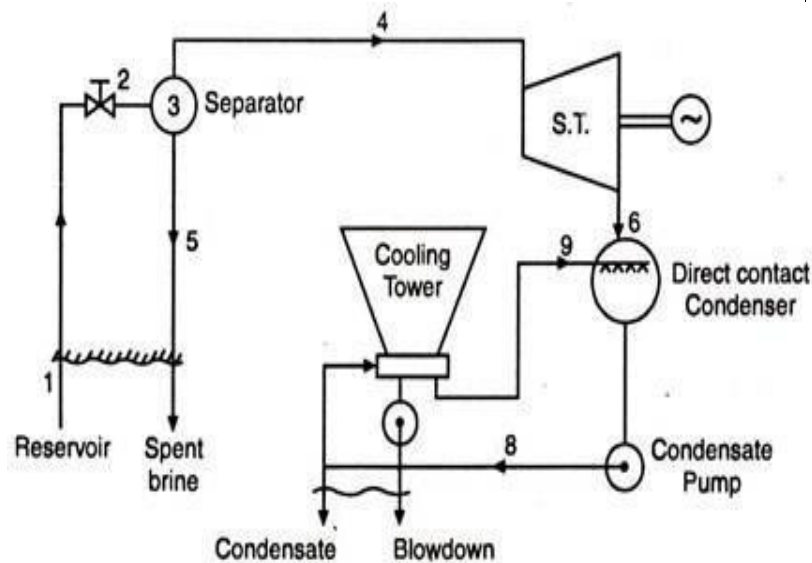


Fig. 7.3. Single-flash steam power plant

The Flashed-Steam System:

Hot water is available above 150°C to 315°C underground. When tapped, the water can flow naturally under its own pressure or be pumped to the surface of the earth. The drop in pressure causes it to partially flash into steam and a liquid-dominated, low-quality, two-phase mixture of water and steam is available at the wellhead. The water contains dissolved solids. Hot water from the reservoir reaches the well head. Pressure P_2 is lower than P_1 and. The two-phase mixture of low quality is passed through a flash separator. The dry saturated steam at pressure of about 8 bars is expanded in the steam turbine. The separated brine is re-injected into the ground.

XIV **Bulb Type Turbine**
Marks

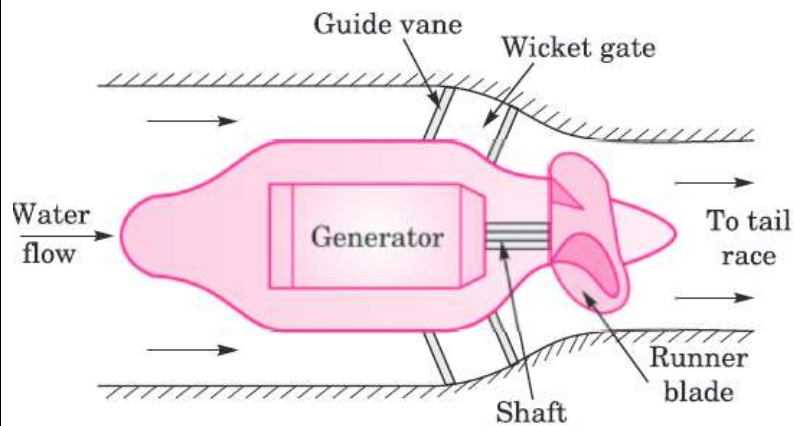
Diagram - 3.5

Description - 3.5 Marks

3.5 +
3.5

7

The bulb turbine is a reaction turbine of Kaplan type which is used for extremely low heads. The characteristic feature of this turbine is that the turbine components as well as the generator are housed inside a bulb, from which the name is developed. The main difference from the Kaplan turbine is that the water flows in a mixed axial-radial direction into the guide vane cascade and not through a scroll casing. The guide vane spindles are normally inclined to 60° in relation to the turbine shaft and thus results in a conical guide vane cascade contrary to other types of turbines. The runner of a bulb turbine may have different numbers of blades depending on the head and water flow. The bulb turbines have higher full-load efficiency and higher flow capacity as compared to Kaplan turbines. It has a relatively lower construction cost. The bulb turbines can be utilized to tap electrical power from the fast flowing rivers on the hills. Figure shows the schematic of a Bulb Turbine Power Plant



Scoring Indicators

Model Question Paper II

RENEWABLE ENERGY ENGINEERING-11

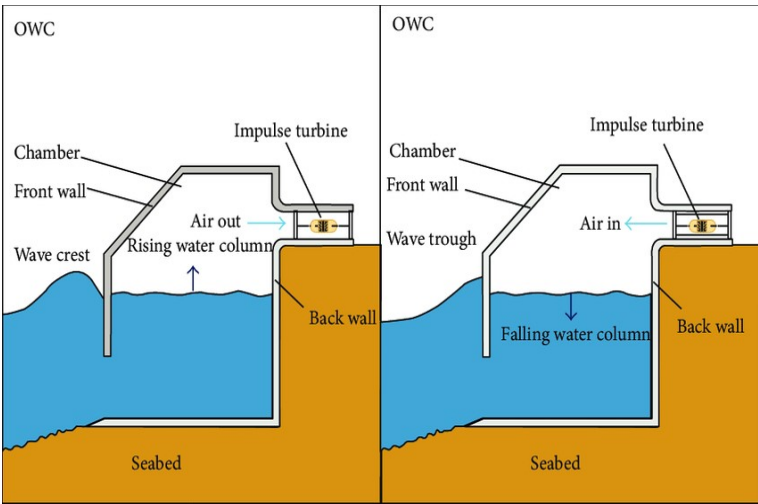
Q No	Scoring Indicators	Split score	Sub Total	Total Score
	PART A			9
I. 1	Schemes for wind power generation <i>list any two</i> <i>2* 0.5 marks = 1 mark</i>	0.5+ 0.5	1	
	● constant speed constant frequency scheme ● variable speed constant frequency scheme			
I. 2	Applications of wind power. <i>list any two</i> <i>2* 0.5 marks = 1 mark</i>	0.5+ 0.5	1	
	wind pumps wind battery charges wind electricity generators			
I. 3	Applications of Fuel cell <i>list-any two</i> <i>2* 0.5 marks = 1 mark</i>	0.5+0.5	1	
	● Fuel cell electric vehicles, or FCEVs, use clean fuels and are therefore more eco-friendly than internal combustion engine-based vehicles. ● They have been used to power many space expeditions including the Apollo space program. ● Generally, the byproducts produced from these cells are heat and water. ● The portability of some fuel cells is extremely useful in some military applications. ● These electrochemical cells can also be used to power several electronic devices. ● Fuel cells are also used as primary or backup sources of electricity in many remote areas.			
I. 4	Ocean Energy Resources <i>0.5mark each</i> <i>2* 0.5 marks = 1 mark</i>	0.5+ 0.5	1	

	1.Tides 2.waves			
I. 5	Definition of Tide <i>1 mark</i>	1	1	
	Tides are the rise and fall of sea levels caused by gravitational forces exerted by the Moon and Sun and the rotation of Earth			
I. 6	Definition of Tidal range <i>1 mark</i>	0.5+0.5	1	
	Tidal range is the height difference between high tide and low tide			
I. 7	limitations of wave energy <i>0.5mark each 2* 0.5 marks = 1 mark</i>	1	1	
	1) Wave energy can only be used in areas where Waves are very strong. 2) Since the waves are erratic, the energy produced from wave energy is unpredictable. So, it is not a reliable source of energy. 3) The initial set-up costs are high. 4) It can harm the marine ecosystem.			
I. 8	Magma Resources <i>1 mark</i>	1	1	
	Magma is an extremely hot liquid and semi-liquid rock located under Earth's surface. This magma can push through holes or cracks in the crust, causing a volcanic eruption. When magma flows or erupts onto Earth's surface, it is called lava . Like solid rock, magma is a mixture of minerals			
I. 9	Water turbines used in SHP <i>Any two 0.5*2 =1</i>	0.5+ 0.5	1	
	Bulb type turbine, Tube type turbine, Kaplan turbine			
	PART B			24

II. 1	Limitations of WECS <i>List any three -Each point 1 mark</i>	3	3	
	• Wind power must still compete with conventional generation sources on a cost basis. Even though the cost of wind power has decreased dramatically in the past several decades, wind projects must be able to compete economically with the lowest-cost source of electricity, and some locations may not be windy enough to be cost competitive. • Good land-based wind sites are often located in remote locations, far from cities where the electricity is needed. Transmission lines must be built to bring the electricity from the wind farm to the city.. • Wind resource development might not be the most profitable use of the land. Land suitable for wind-turbine installation must compete with alternative uses for the land, which might be more highly valued than electricity generation. • Turbines might cause noise and aesthetic pollution. • Wind plants can impact local wildlife.			
II. 2	Classification of WECS <i>Total-3marks</i>	3	3	
	1) Based on axis (a) Horizontal axis machines (b) Vertical axis machines (2) According to size (a) Small size machines (up to 2 KW) (b) Medium size machines (2 to 100 KW) (c) Large size machines (100KW and above)			
	(3) According to the rotational speed of the aero turbines (1) Constant speed and variable pitch blades (2) Nearly constant speed with fixed pitch blades (3) Variable speed with fixed pitch blades (4) As per utilization of output (a) Battery storage (b) Direct conversion to an electromagnetic energy converter (c) Thermal potential (d) Inter convention with conventional electric utility guides			

II. 3	Difference between Battery and Fuel cell <i>Total -3 marks</i>	3	3	
	The key difference between fuel cell and battery is that a fuel cell can supply electrical energy for a long period of time compared to a normal battery. Furthermore, a fuel cell is continuously supplied with fuel and oxygen from an external source, which makes it work for a long time period; however, a battery contains a limited amount of fuel and oxidant, and these two components decrease with time, so this device cannot supply electrical energy for a long period of time.			
II. 4	Basic principle of OTEC <i>Total -3 marks</i>	3	3	
	OTEC or ocean thermal energy thermal conversion is a technology which converts solar radiation absorbed by the oceans to electric energy. The working principle of an OTEC plant is that it uses the warm water to heat and vaporize a liquid working fluid. And this working fluid develops pressure which forces it to evaporate. and the expanding vapor runs through a heat engine like turbine, generator, and it is condensed back into a liquid by cold water brought up from depth and the cycle is repeated.			
II. 5	Block Diagram of a Hybrid cycle OTEC system <i>Total -3 marks</i>	3	3	

II. 6	High level Reservoir Wave Machine <i>Figure-1.5 marks, explanation -1.5 marks</i> High-Level Reservoir Wave Machine The concept of this device is illustrated with reference to Fig. 9.17, in which a magnification piston is used. The pressurized water is elevated to a natural reservoir above the wave generator, which would have to be near a shore line, or to an artificial water reservoir. The water in the reservoir is made to flow through a turbine coupled to an electric generator, and then back to sea level. Calculations made shown that a 20 m diameter generator can produce 1 MW power. Fig. 9.17. Schematic of a high level reservoir wave machine.	1.5+1.5 5	3	
II. 7	Schematic layout <i>figure-3 marks</i>	3	3	

				
II. 8	Nature of geothermal fields <i>Total-3 marks</i>	3	3	
	It is convenient to classify earth's surface into three broad groups. (1) <i>Nonthermal areas</i> having a temperature gradient of 10–40°C per km depth. (2) <i>Semithermal areas</i> having a temperature gradient of 70°C per km depth. (3) <i>Hyperthermal areas</i> where the temperature gradients are many times greater than in non thermal areas. Geothermal fields may further be classified into three types: (A) Hyperthermal Fields (1) <i>Wet fields</i>. Where the water is pressurized and temperatures are above 100°C. When they are led to the surface a fraction will be splashed into steam and a major part remains as the boiling water. (2) <i>Dry fields</i>. They produce dry saturated steam or superheated steam at pressure above atmospheric. (B) Semithermal Fields These are capable of producing hot water at temperatures above 100°C.			
II. 9	Kaplan Turbine <i>explanation:3 marks</i>	3	3	

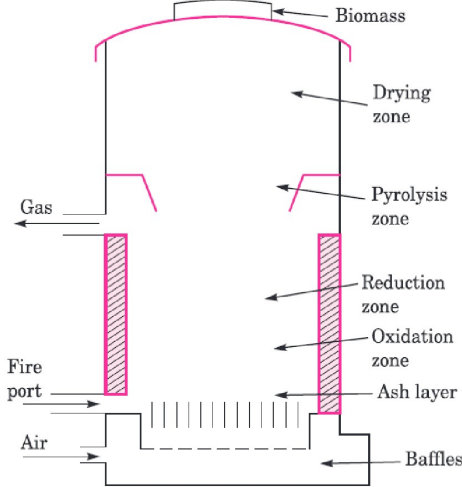
	It is an axial flow reaction turbine where water enters and leaves the runner vanes parallel to the axis of the shaft. It is particularly suited for low head up to 30 m and high discharge of water. The main components of Kaplan turbine are 1. Scroll casing 2. Guide vane mechanism 3. Boss with adjustable vanes 4. Draft tube			
	Since Kaplan and Francis turbines are reaction turbines, the working principle of the Kaplan turbine is the same as that of the Francis turbine. Water from the penstock flows into the scroll casing surrounding the turbine runner. From the scroll casing water flows through the guide mechanism and enters the runner vanes. The force exerted by water on the vanes causes the runner shaft to rotate. After imparting energy to the runner, the water is discharged through the draft tube.			
II. 10	Hot Dry Rock Resources <i>explanation: 3 marks</i>	3	3	
	Hot dry rock is made up of granite. It is a unique geothermal resource that lacks stored fluid and has extremely low permeability. Fracture creation is required for commercial applications. Hot dry rock (HDR) is an extremely abundant source of geothermal energy that is difficult to access. A vast store of thermal energy is contained within hot – but essentially dry and impervious crystalline basement rocks found almost everywhere deep beneath Earth's surface. Naturally occurring geothermal energy is stored in hot rocks below the surface of the Earth. Water moving through fractures in the rocks becomes steam that can be delivered via geothermal wells to the surface. Once at the surface, steam moves through pipes into the power plant and drives a turbine and generator system. Hot dry rock (HDR) systems require such rocks to be artificially fractured at depth so that water can be pumped into them, heated above boiling temperature and then returned to the surface to flash to steam in an electricity generating plant.			
	PART C			42

III	Wind power calculation <i>Area calculation: 2 marks</i> <i>Wind power calculation: 5 marks.</i>	2+5	7	
	• Wind speed $v = 20$ m/s, • Blade length $l = 50$ m, • Air density $\rho = 1.23$ kg/m. • The area is given by, $A = \pi r^2$ • $A = \pi \times 2500$ • $= 7850$ m • The wind power formula is given as, • $P = \frac{1}{2} \rho A V^3$ • $P = \frac{1}{2} (1.23) (7850) 20^3$ • P = 38622 W			
IV	Power available from the wind <i>Derivation :7 marks</i>	7	7	

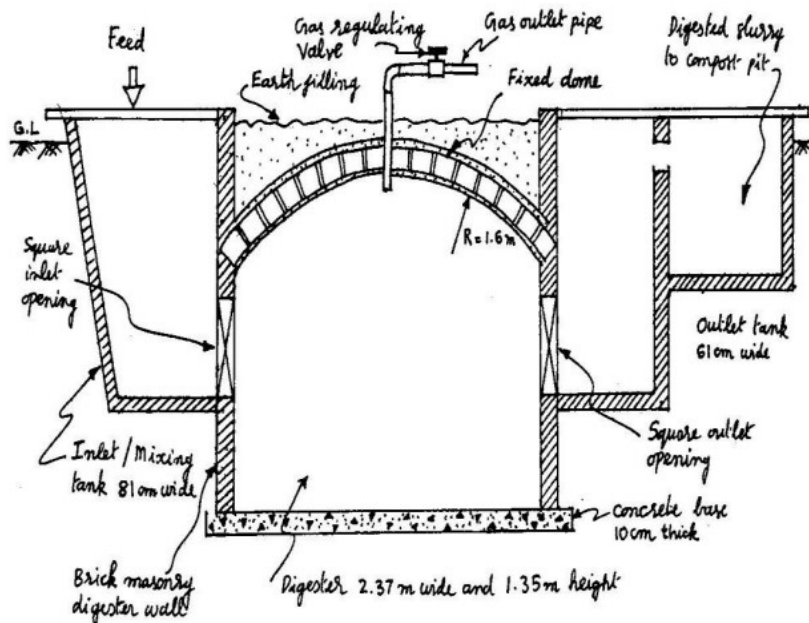
	Wind possesses energy by virtue of its motion. Any device capable of slowing down the mass of moving air, like a sail or propeller, can extract part of the energy and convert it into useful work. Three factors determine the output from a wind energy converter: (i) the wind speed; (ii) the cross-section of wind swept by rotor; and (iii) the overall conversion efficiency of the rotor, transmission system and generator or pump. The power in the wind can be computed by using the concept of kinetics. The wind mill works on the principle of converting kinetic energy of the wind to mechanical energy. We know that power is equal to energy per unit time. The energy available is the kinetic energy of the wind. The kinetic energy of any particle is equal to one half its mass times the square of its velocity, or $\frac{1}{2} mV^2$. The amount of air passing in unit time, through an area A, with velocity V, is $A \cdot V$, and its mass m is equal to its volume multiplied by its density ρ of air, or $m = \rho AV \quad \dots(6.1)$ $= \frac{1}{2} \rho AV^3 \text{ watts} \quad \dots(6.2)$ Equation (6.2) tells us that the maximum wind available the actual amount will be somewhat less because all the available energy is not extractable—is proportional to the cube of the wind speed. It is thus evident that small increase in wind speed can have a marked effect on the power in the wind. Equation (6.2) also tell us that the power available is proportional to air density (1.225 kg/m³ at sea level). It may vary 10-15 % during the year because of pressure and temperature change. It changes negligibly with water content. Equation also tells us that the wind power is proportional to the intercept area. Thus an aeroturbine C with a large swept area has higher power than a smaller area machine; but there are added implications. Since the area is normally circular of diameter D in <i>horizontal axis</i> aeroturbines, then $A = \frac{\pi}{4} D^2$, (sq. m.), which when put in equation (6.2) gives, $\begin{aligned} \text{Available wind power } P_a &= \frac{1}{2} \rho \frac{\pi}{4} D^2 V^3 \text{ watts} \\ &= \frac{1}{8} \rho \pi D^2 V^3 \quad \dots(6.3) \end{aligned}$			
V	<u>Comparison</u> <i>comparison: 7marks</i>	7	7	

	H o r i z o n t a l	V e r t i c a l
	A x i s i s V i n d T u r b i n e	A x i s V i n d T u r b i n e
	A x i s o f r o t a t i o n i	A x i s o f r o t a t i o n i

VI	Characteristics and advantages of Darrieus Rotor <i>characteristics : 3.5marks</i> <i>advantages: 3.5marks</i>	3.5+3 .5	7	
	Characteristics of Darrieus Rotor (i) Note self starting (ii) High speed (iii) High efficiency (iv) Potentially low capital cost. Advantages. Advantages of such WEC system are: (1) The major advantage of this design is that the rotor blades can accept the wind from any compass. (2) Another added advantage is that the machine can be mounted on the ground eliminating tower structures and lifting of huge weight of machine assembly, i.e., it can be operated close to the ground level. (3) Since this machine has vertical axis symmetry, it eliminates yaw control requirement for its rotor to capture wind energy. A dual purpose and relatively simple shaft axis support is anticipated as well as ground level power output delivery due to presence of vertical shaft. This may in turn, allow easier access and serviceability. (4) Airfoil rotor fabrication costs are expected to be reduced over conventional rotor blade costs. (5) The absence of pitch control requirements for synchronous operation may yield additional cost savings. (6) The tip speed ratio and power coefficient are considerably better than those of the S-rotor but are still below the values for a modern horizontal-axis, two-bladed propeller rotor.			
VII	Updraft Type biomass gasifiers <i>Figure-3.5 marks</i> <i>Explanation: 3.5 mark</i>	3.5+3 .5	7	

	<i>(i) Up Draught Gasifiers.</i> In updraught gasifier air enters below the combustion zone and the producer gas leaves near the top of the gasifier. This type of gasifier is easy to build and operate. The gas produced has practically no ash but contains tar and water vapour because of passing of gas through the unburnt fuel. Hence updraught gasifiers are suitable for tar free fuels like charcoal, especially in stationary engines (Fig. 7.26).  Fig. 7.26. Schematic diagram of an updraught gasifier.			
VIII	JANATA model of Biogas plant <i>Diagram: 3.5 marks</i> <i>Description: 3.5 marks</i>	3.5+ 3.5	7	
	The design of this plant is of Chinese origin but it has been introduced under the name “Janata biogas plant” by Gobar Gas Research Station, Ajitmal in view of its reduced cost. This is a plant where no steel is used, there is no moving part in it and maintenance cost is low. The plant can be constructed by village mason taking some pre-explained precautions and using all the indigenously available building materials. Good quality bricks and cement should be used to avoid the afterward structural problems like cracking of the dome and leakage of gas.			

This model has a higher capacity when compared with the KVIC model, hence it can be used as a community biogas plant. This design has a longer life than KVIC models. Substrates other than cattle dung such as municipal waste and plant residues can also be used in janata type plants. The plant consists of an underground well sort of digester made of bricks and cement having a dome shaped roof which remains below the ground level is shown in figure. At almost the middle of the digester, there are two rectangular openings facing each other and coming up to a little above the ground level, acting as an inlet and outlet of the plant. Dome shaped roof is fitted with a pipe at its top which is the gas outlet of the plant. The biogas is collected in the restricted space of the fixed dome, hence the pressure of gas is much higher, which is around 90 cm of water column



IX

Pyrolysis and Liquefaction process

explanation: 3.5 marks each

3.5+3
.5

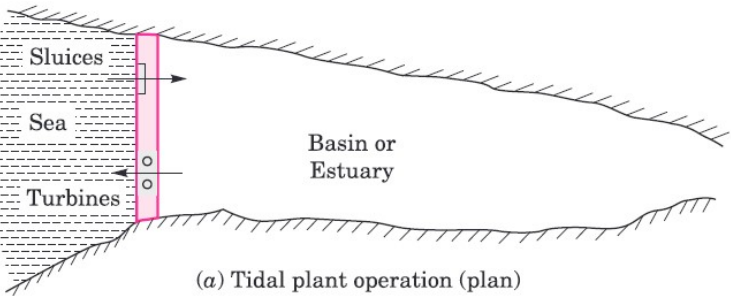
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	<i>Pyrolysis.</i> A wide range of energy rich fuels can be produced by roasting dry woody matter like straw and wood chips. The material is fed into a reactor vessel or retort in a pulverised or shredded form and heated in the absence of air. (Air would cause the products of pyrolysis to ignite). As the temperature rises the cellulose and lignin break down to simpler substances which are driven off leaving a char residue behind. <u>This process has been used for centuries to produce charcoal.</u> <i>Liquefaction.</i> Liquid yields are maximized by rapid heating of the feedstock to comparatively low temperatures. The vapours are condensed from the gas stream and these separate into a two phase liquor: the aqueous phase (pyroligneous acid) contains a soup of water-soluble organic materials like acetic acid, acetone and methanol ('wood alcohol'); the non-aqueous phase consists of oils and tars. These crude products can be burnt (with some difficulty), but it is usually more profitable to up-grade them to premium fuels by conventional refining techniques.			
X	Continuous and Batch type Biogas Plant <i>explanation: 3.5 marks each</i>	3.5+3.5	7	

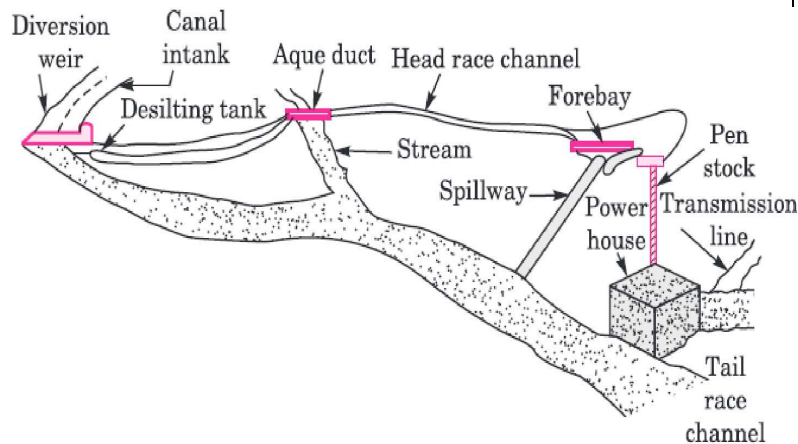
	• (a) Continuous plant- There is a single digester in which raw materials are charged regularly and the process goes on without interruption except for repair and cleaning etc. In this case the raw material is self buffered (like cow dung) or otherwise thoroughly mixed with the digesting mass where dilution prevents souring and the biogas production is maintained. The continuous process may be completed in a single stage or separated into two stages. (i) Single stage process. The entire process of conversion of complex organic compounds into biogas is completed in a single chamber. This chamber is regularly fed with the raw materials while the spent residue keeps moving out. Serious problems are encountered with agricultural residues where fermented in a single stage continuous process (ii) Double stage process. The acidogenic stage and methanogenic stage are physically separated into two chambers, Thus the first stage of acid production is carried out in a separate chamber and only the diluted acids are fed into the second chamber where biomethanation takes place and the biogas can be collected from the' second chamber.			
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	(b) The Batch Plant- The feeding is between intervals, the plant is emptied once the process of digestion is complete. In this type, a battery of digesters are charged along with lime, urea etc. and allowed to produce gas for 40- 50 days. These are charged and emptied one by one in a synchronous manner which maintains a regular supply of the gas through a common gas holder. Sometimes the freshly charged digester is aerated for a few days after which it is' closed to atmosphere. The biogas supply may be utilized after 8-10 days. Obviously such a plant would be expensive to install and unless operated on a large scale it would not be economical. Such systems have been generally installed in European countries. Their installation and operation being capital and labor intensive. They are totally unsuitable for Indian conditions. Except when it is taken as a commercial venture.			
XI	Open cycle OTEC system. <i>Diagram: 3.5 marks</i> <i>Description: 3.5 marks</i>	3.5+ 3.5	7	

	In this system, the warm water from the ocean surface is fed to the deaerator. It removes the dissolved non-condensable gasses from water and feeds it to the evaporator. • The flash evaporator is maintained under high vacuum. As a result, a low-pressure steam is produced due to throttling effect and the residue liquid is released back to the ocean at high depth. • In the next step, this low-pressure steam is fed to the turbine where it expands and rotates the turbine. The mechanical power so developed is converted into electrical power by the generator. • The exhaust steam from the turbine is released into a direct contact type condenser where it mixes with the cold water drawn from the ocean depth. It condenses the exhaust steam and the mixture is discharged into the ocean.			
XII	Single Basin arrangement <i>Diagram: 3.5 marks</i> <i>Description: 3.5 marks</i>	3.5+ 3.5	7	

	A tidal power plant based on a single basin system is shown in the figure. The three main components are power house, Dam to form basin, sluice ways from the basin to the sea and vice versa. The function of the dam is to form a barrier between the sea and the basin. The sluice ways are used either to fill the basin during flood tide or empty the basin during a ebb tide. In this system the water is stored in the tidal basin during flood tide by opening the sluice ways. During the ebb tide the water is allowed to flow through the turbine which generates mechanical power. The turbine is connected to a generator which converts mechanical power into electrical power. The main disadvantage of this system is its operation is intermittent.  (a) Tidal plant operation (plan)			
XIII	Liquid dominated Binary cycle Geothermal power plant <i>Diagram-- 3.5 marks</i> <i>Explanation- 3.5 marks</i>	3.5 + 3.5	7	

	Water in lower temperature ranges is unsuitable for power production. It can be used directly for domestic and industrial process heating. Hot water can also be used in heating an organic fluid with low boiling point and can be used to run a Rankine Cycle. The working fluids can be isobutene, Freon-12, ammonia or propane. Hot brine from underground reservoirs circulates through a heat exchanger (HX) and is pumped back to ground. The organic fluid is heated to superheated vapor and is used in a standard closed Rankine cycle. The vapor drives the turbine and is condensed in a surface condenser. The condenser is cooled by water from a natural source or a cooling tower circulation system. There is no problem of corrosion and scaling in the working cycle components. Fig. 7.7. Binary Cycle System			
XIV	Layout of SHP <i>Diagram - 3.5 Marks</i> <i>Description - 3.5 Marks</i>	3.5 + 3.5	7	



- The basic components are Diversion and intake, Desilting chamber, Water conductor system, fore bay, Penstock, Power house and Tail race channel.
- Dam, Barrages, solid boulder structure and trench type weir are usually employed to divert the required flow from the river bed /streams to the intake structure. The intake is brought through a canal and then for bay. The fore bay is used for sediment holding. At the bottom of the system the water is tunneled through a pipeline (penstock) to the powerhouse building containing a turbine. The penstock builds up pressure from the water that has traveled downwards. In mountainous areas, access to the route of the penstock may provide considerable challenges. If the water source and turbine are far apart, the construction of the penstock may be the largest part of the costs of construction. At the turbine, a controlling valve is installed to regulate the flow and the speed of the turbine. The turbine converts the flow and pressure of the water to mechanical energy; the water emerging from the turbine returns to the natural watercourse along a tailrace channel. The turbine runs a generator, which is then connected to electrical loads; this might be directly connected to the power system of a single building in very small installations, or may be connected to a community distribution system for several homes or buildings.